

```
message")
st.warning("This is a Warning message")
st.error("This is an Error message ")
st.markdown('Streamlit is an Awesome
Library'.')
'''
st.code(code, language='python')
```

Using a DataFrame and chart

Streamlit enables the usage of data frames and building multiple types of charts with direct functions. A sample code snippet along with the output is shown in Figure 8:

```
chart_data = pd.DataFrame(
    np.random.randn(20, 3),
    columns=['Team I', 'Team II', 'Team
III'])
```

```
st.area_chart(chart_data)
st.bar_chart(chart_data)
```

Images and other media

Streamlit enables inclusion of images and other media such as audio and video in the interface (Figure 9):

```
from PIL import Image
image = Image.open('mywindow.jpeg')
st.image(image, caption='A Rainy Day
at Pondicherry University', use_column_
width=True)
```

```
audio_file = open('isai.ogg', 'rb')
audio_bytes = audio_file.read()
st.audio(audio_bytes, format='audio/ogg')
```

Interactive widgets

Widgets are the most important user interface components. Streamlit has various widgets, which are explained in this section.

To add an interactive button, type:

```
if st.button("Click Me"):
    st.success("Successfully Completed")
```

A button with the label 'Click Me' is included (Figure 10). When the user clicks the button, the message 'Successfully Completed' appears with the success



Figure 9: Inclusion of images and other media

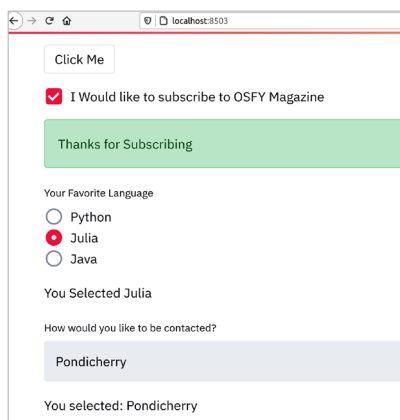


Figure 10: Button, CheckBox, Radio and SelectBox

notification explained earlier.

To add an interactive CheckBox, type:

```
sub = st.checkbox('I Would like to
subscribe to OSFY Magazine')
if sub:
    st.write('Thanks for Subscribing')
```

To add an interactive radio button, write the following code:

```
lang = st.radio( "Your Favourite
Language", ('Python', 'Julia',
'Java'))
if lang == 'Python':
    st.write('You selected Python.')
if lang == 'Julia':
```

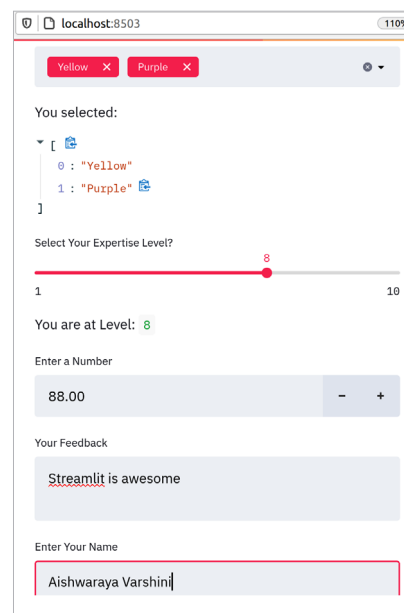


Figure 11: Multiselect, slider, number input, text input and text area

```
st.write("You Selected Julia")
if lang == 'Java':
    st.write("You Selected Java")
```

To add a Select box, type:

```
city = st.selectbox('How would you
like to be contacted?', ('Chennai',
'Bangalore', 'New Delhi', 'Mumbai',
'Kolkata', 'Pondicherry'))
st.write('You selected:', city)
```